MAT A22 — Linear Algebra I for Mathematical Sciences Toolkit

Winter 2024

This is a compilation of the theorems and definitions covered in the course notes. The weeks are numbered the same as the course notes, and the theorems are numbered for cross-referencing purposes.

Credits to Professor Parker Glynn-Adey and Professor Thomas Kielstra for the notes from which this fact compilation was created.

Contents

1	Vector Spaces	5
2	Linear Combinations	9
3	Systems of Linear Equations	11
4	Bases and Dimension	14
5	Linear Transformations	16
6	Kernel and Image	20
7	Applications of the Dimension Theorem	21
8	Composition and Inverses	23
9	Change of Basis and Determinants	26
10	Eigenvalues and Eigenvectors	28
11	Diagonalizability	29

Fact List

1.1	Binary Operations	5
1.2	Commutative and Associative	5
1.3	The Axioms of a Field	5
1.4	The Imaginary Number i	5
1.5	Complex Numbers	6
1.6	Subtracting Complex Numbers	6
1.7	Complex Conjugate	6
1.8	Modulus of a Complex Number	6
1.9	Dividing Complex Numbers	6
1.10	The Axioms of a Vector Space	6
1.11	Complex Vector Spaces	7

1.12 Real Vector Spaces . . . 7
1.13 The Vector Space $\mathbb{R}^n$ . . . 7
1.14 The Vector Space of Functions $\mathbf{F}(\mathbb{R})$ . . . 7
1.15 The Vector Space of $n \times k$ Matrices $\mathbf{M}_{n\times k}(\mathbb{R})$ . . . 7
1.16 Uniqueness of the Zero Vector . . . 7
1.17 Scaling by Zero . . . 8
1.18 Additive Inverse and Scaling . . . 8
1.19 Uniqueness of the Additive Inverse . . . 8
1.20 Subspaces . . . 8
1.21 Characterization of Subspaces by Closure . . . 8
1.22 Subspaces Closed Under Intersection . . . 8
2.1 Linear Combinations . . . 9
2.2 Spans . . . 9
2.3 Spans Are Subspaces . . . 9
2.4 Sums of Subsets . . . 9
2.5 Sums of Subspaces . . . 9
2.6 Sums of Subspaces . . . 9
2.7 Redundant Vector . . . 9
2.8 Linear Dependence and Linear Independence . . . 9
2.9 Redundant Vectors and Linear Independence . . . 9
2.10 Independence and Uniqueness of Linear Combinations . . . 10
3.1 Linear Systems . . . 11
3.2 Homogeneous Systems . . . 11
3.3 The Trivial Solution . . . 11
3.4 Augmented Matrix Notation for Linear Systems . . . 11
3.5 The Row Operations . . . 12
3.6 Row Operations Preserve the Solution Sets of Linear Systems . . . 12
3.7 Echelon Form . . . 13
3.8 The Elimination Algorithm . . . 13
3.9 Every Linear System Has a Row Echelon Form . . . 13
3.10 Basic and Free Variables . . . 13
3.11 Solutions of Homogeneous Systems and Parameters . . . 13
3.12 Affine Sets . . . 13
4.1 Spanning Sets and Bases . . . 14
4.2 Characterizing Bases by Unique Representation . . . 14
4.3 The Linear Independence Extension Lemma . . . 14
4.4 Extending a Linearly Independent Set to a Basis . . . 14
4.5 The Dimension Bound . . . 14
4.6 All Bases are the Same Size . . . 14
4.7 Dimension . . . 14
4.8 Independence and Cardinality . . . 14
4.9 Proper Subspace Dimension . . . 15
5.1 Linear Transformation . . . 16
5.2 Linear Transformations Preserve Additive Identities . . . 16
5.3 Linear Transformations Preserve Addition and Scaling . . . 16
5.4 Zero Transformation . . . 16
5.5 Identity Transformation . . . 16

5.6 Linearity of Zero and Identity Transformations . . . 16
5.7 The Standard Inner Product . . . 16
5.8 Properties of the Standard Inner Product Over $\mathbb{R}^n$ . . . 17
5.9 Inner Products Define Angles . . . 17
5.10 Orthogonality . . . 17
5.11 Rotation in $\mathbb{R}^2$ . . . 17
5.12 Projection in $\mathbb{R}^2$ . . . 17
5.13 Linear Maps are Determined by Their Actions on a Basis . . . 17
5.14 $T\colon V \to W$ Characterized by $\dim(V) \times \dim(W)$ Numbers . . . 17
5.15 The Matrix $[T]_\alpha^\beta$ of a Linear Transformation . . . 18
5.16 Coordinate Vectors . . . 18
5.17 Matrix-Vector Multiplication . . . 18
5.18 Matrix-Vector Multiplication in Coordinates $[T(\mathbf{v})]_\beta = [T]_\alpha^\beta[\mathbf{v}]_\alpha$ . . . 18
5.19 The Transformation Defined by a Matrix . . . 19
5.20 The Fundamental Theorem of Matrices . . . 19
6.1 Kernel . . . 20
6.2 The Kernel is a Subspace . . . 20
6.3 Image . . . 20
6.4 The Image is a Subspace . . . 20
6.5 Kernels in Coordinates . . . 20
6.6 Image and Spans . . . 20
6.7 Image and Columns ("Procedure 1") . . . 20
6.8 Bases for Image and Kernel ("Procedure 2") . . . 20
6.9 The Rank-Nullity Theorem . . . 20
7.1 Injective and Surjective . . . 21
7.2 Kernels and Injectivity . . . 21
7.3 Image and Surjectivity . . . 21
7.4 Dimension and Injectivity . . . 21
7.5 Dimension and Surjectivity . . . 21
7.6 In Finite Dimensions Injectivity and Surjectivity are Equivalent . . . 21
7.7 Inverse Sets . . . 21
7.8 Affine Subsets . . . 22
7.9 Solutions Sets of Linear Systems . . . 22
7.10 Solutions Sets of Linear Systems and Kernels . . . 22
8.1 Composition . . . 23
8.2 A Composition of Linear Maps is Linear . . . 23
8.3 Addition of Linear Maps . . . 23
8.4 Composition Has Nice Properties . . . 23
8.5 Kernels, Images, and Composition . . . 23
8.6 The Matrix Product . . . 23
8.7 The Matrix Multiplication Formula . . . 24
8.8 An Inverse of a Function . . . 24
8.9 Injectivity, Surjectivity, and Inverses . . . 24
8.10 Inverses are Unique . . . 24
8.11 The Inverse is Linear . . . 24
8.12 Elementary Row Operations are Invertible Matrices . . . 24
8.13 Isomorphism . . . 24

8.14 Endomorphism . . . 24
8.15 Automorphism . . . 24
8.16 Sets of Linear Transformations . . . 24
8.17 The Automorphism Group of a Vector Space . . . 25
8.18 The Gauss-Jordan Algorithm for Finding Inverses . . . 25
8.19 Isomorphisms and Injectivity/Surjectivity . . . 25
8.20 Isomorphic . . . 25
8.21 The Classification of Finite Dimensional Vector Spaces . . . 25
8.22 Real Vector Spaces Isomorphism and Dimensions . . . 25
9.1 Alternating and Multilinear . . . 26
9.2 Alternating and Repeated Entries . . . 26
9.3 Alternating, Multilinear, and Dependent Input Set . . . 26
9.4 The Characterization of the det Function . . . 26
9.5 Diagonal Matrices . . . 26
9.6 Determinants and Row Operations . . . 26
9.7 Determinants and Invertibility . . . 26
9.8 The 1×1 and 2×2 Determinants . . . 27
9.9 Minors (Submatrices) . . . 27
9.10 The Recursive Formula for det . . . 27
9.11 Changing Coordinates . . . 27
9.12 Changing Coordinates and Linear Maps . . . 27
9.13 A Useful Special Case: Changing Coordinates of $T \colon V \to V$ . . . 27
10.1 n-fold Composition . . . 28
10.2 Eigenvalues and Eigenvectors . . . 28
10.3 Eigenvectors Are Vectors in Kernels . . . 28
10.4 The Characteristic Polynomial . . . 28
10.5 Eigenvalues are Roots of the Characteristic Polynomial . . . 28
10.6 The λ-Eigenspace E_λ . . . 28
10.7 E_λ is a Subspace . . . 28
10.8 Determinants and Eigenvalues . . . 28
10.9 Invertibility and Eigenvalues . . . 28
11.1 The LADR Approach to Finding Eigenvectors: Find a Dependence . . . 29
11.2 The LADR Approach: Factor and Apply the Dependence . . . 29
11.3 The Fundamental Theorem of Algebra . . . 29
11.4 A Bound on The Number of Eigenvalues . . . 29
11.5 Diagonalizable . . . 29
11.6 Characterization of Diagonalizable . . . 29
11.7 Eigenvectors of Distinct Eigenvalues are Independent . . . 29
11.8 The Nice Case of Diagonalization . . . 30
11.9 Algebraic and Geometric Multiplicity . . . 30
11.10 Geometry $\leq$ Algebra . . . 30
11.11 Independent Sets from Eigenspaces . . . 30
11.12 The General Case of Diagonalization . . . 30

1 Vector Spaces

Definition 1.1 — Binary Operations

A binary operation is a function $\boxplus\colon X \times X \to X$ with two inputs.

- The usual addition on real numbers $+\colon \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ is a good example.

$$2 + 3 = 3 + 2 = 5$$

- Another common example is the usual multiplication $\cdot\colon \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ of real numbers.

$$3 \cdot 2 = 3 \cdot 2 = 6$$

Definition 1.2 — Commutative and Associative

An operation $\boxplus\colon X \times X \to X$ is *commutative* if:

$$x \boxplus y = y \boxplus x$$

An operation $\boxplus\colon X \times X \to X$ is *associative* if:

$$x \boxplus (y \boxplus z) = (x \boxplus y) \boxplus z$$

Notation: We use the notation $\boxplus$ for a general operation with two inputs.

Definition 1.3 — The Axioms of a Field

A *field* is a set $\mathbb{F}$ together with two binary operations

$$\boxplus\colon \mathbb{F} \times \mathbb{F} \to \mathbb{F} \qquad\qquad \boxdot\colon \mathbb{F} \times \mathbb{F} \to \mathbb{F}$$

called *addition* and *multiplication* such that the following axioms hold:

A1. $\boxplus$ and $\boxdot$ are both associative

A2. $\boxplus$ and $\boxdot$ are both commutative

A3. There exists an additive identity $\hat{0} \in \mathbb{F}$ and a multiplicative identity $1 \in \mathbb{F}$ such that:

$$\hat{0} \boxplus x = x \qquad\qquad 1 \boxdot x = x$$

for all $x \in \mathbb{F}$.

A4. For each $x \in \mathbb{F}$, there is an additive inverse $x' \in \mathbb{F}$ such that:

$$x \boxplus x' = \hat{0}$$

A5. For each $x \in \mathbb{F}$ with $x \neq \hat{0}$, there is a multiplicative inverse $x' \in \mathbb{F}$ such that:

$$x \boxdot x' = 1$$

A6. The multiplication operator is distributive over the addition operator:

$$a \boxdot (b \boxplus c) = a \boxdot b \boxplus a \boxdot c$$

Definition 1.4 — The Imaginary Number i

The *imaginary number* i is defined as follows:

$$i = \sqrt{-1}$$

Definition 1.5 — Complex Numbers

Complex numbers are of the form

$$z = a + bi$$

where $a, b \in \mathbb{R}$. Here, $a = \mathrm{Re}(z)$ is the *real part* and $b = \mathrm{Im}(z)$ is the *imaginary part* of the complex number z. The *set of complex numbers* is denoted by $\mathbb{C}$.

Let $z_1 = a + bi$ and $z_2 = c + di$ be complex numbers (where $a, b, c, d \in \mathbb{R}$). *Addition* over $\mathbb{C}$ is defined as

$$z_1 + z_2 = (a + c) + (b + d)i$$

and *multiplication* is defined as

$$z_1 z_2 = (a + bi)(c + di) = (ac - bd) + (ad + bc)i$$

Definition 1.6 — Subtracting Complex Numbers

Let $z_1 = a + bi$ and $z_2 = c + di$ be complex numbers. *Subtraction* over $\mathbb{C}$ is as follows:

$$z_1 - z_2 = (a - c) + (b - d)i$$

Definition 1.7 — Complex Conjugate

Let $z = a + bi \in \mathbb{C}$. The *complex conjugate* of z, denoted z^* or $\overline{z}$, is defined as

$$\overline{z} = a - bi$$

Definition 1.8 — Modulus of a Complex Number

The *modulus* of $z \in \mathbb{C}$ is defined as

$$|z| = \sqrt{\mathrm{Re}(z)^2 + \mathrm{Im}(z)^2}$$

Definition 1.9 — Dividing Complex Numbers

Let $z_1, z_2 \in \mathbb{C}$ with $z_2 \neq 0$. *Division* over $\mathbb{C}$ is as follows:

$$\frac{z_1}{z_2} = \frac{z_1 \overline{z_2}}{|z_2|^2}$$

Definition 1.10 — The Axioms of a Vector Space

A vector space is a set V with the binary operations of vector addition $\boxplus\colon V \times V \to V$ and scalar multiplication $\boxdot\colon \mathbb{F} \times V \to V$, where the following axioms hold for all $\mathbf{x}, \mathbf{y}, \mathbf{z} \in V$ and $c, d \in \mathbb{F}$:

A1. $\mathbf{x} \boxplus (\mathbf{y} + \mathbf{z}) = (\mathbf{x} \boxplus \mathbf{y}) + \mathbf{z}$ ($\boxplus$ is *associative*)

A2. $\mathbf{x} \boxplus \mathbf{y} = \mathbf{y} \boxplus \mathbf{x}$ ($\boxplus$ is *commutative*)

A3. There exists a *zero vector* $\mathbf{0} \in V$ such that for all $\mathbf{x} \in V$, $\mathbf{0} \boxplus \mathbf{x} = \mathbf{x}$.

A4. For all $\mathbf{x} \in V$, there exists an *additive inverse* $\mathbf{x}' \in V$ such that $\mathbf{x} \boxplus \mathbf{x}' = \mathbf{0}$.

A5. $c \boxdot (\mathbf{x} \boxplus \mathbf{y}) = (c \boxdot \mathbf{x}) \boxplus (c \boxdot \mathbf{y})$ (scaling *distributes* over addition)

A6. $(c + d) \boxdot \mathbf{x} = (c \boxdot \mathbf{x}) \boxplus (d \boxdot \mathbf{x})$ (field addition in $\mathbb{F}$ *distributes* over scalar multiplication in V)

A7. $c \boxdot (d \boxdot \mathbf{x}) = (cd) \boxdot \mathbf{x}$ ($\boxdot$ *associates* with scalar multiplication in $\mathbb{F}$)

A8. For all $\mathbf{x} \in V$, $1 \boxdot \mathbf{x} = \mathbf{x}$ (1 is the *multiplicative identity* from $\mathbb{F}$)

These axioms have been adapted from Little and Damiano p.7 to use $\boxplus$ and $\boxdot$.

Definition 1.11 — Complex Vector Spaces

A *complex vector space*, or $\mathbb{C}$-vector space, is a vector space where the field is the set of complex numbers.

Definition 1.12 — Real Vector Spaces

A *real vector space*, or $\mathbb{R}$-vector space, is a vector space where the field is the set of real numbers.

Definition 1.13 — The Vector Space $\mathbb{R}^n$

The vector space of *real n-tuples* is given as follows

$$\mathbb{R}^n = \{(x_1, x_2, \ldots, x_n) \colon x_i \in \mathbb{R}\}$$

The addition on $\mathbb{R}^n$ is:

$$(x_1, x_2, \ldots, x_n) + (y_1, y_2, \ldots, y_n) = (x_1 + y_1, x_2 + y_2, \ldots, x_n + y_n)$$

The scalar multiplication on $\mathbb{R}^n$ is:

$$c(x_1, x_2, \ldots, x_n) = (cx_1, cx_2, \ldots, cx_n)$$

Definition 1.14 — The Vector Space of Functions $\mathbf{F}(\mathbb{R})$

The vector space of *real valued functions* is given as follows:

$$\mathbf{F}(\mathbb{R}) = \{f \colon \mathbb{R} \to \mathbb{R}\}$$

The addition on $\mathbf{F}(\mathbb{R})$ is:

$$(f + g)(x) = f(x) + g(x)$$

The scalar multiplication on $\mathbf{F}(\mathbb{R})$ is:

$$(cf)(x) = cf(x)$$

Definition 1.15 — The Vector Space of $n \times k$ Matrices $\mathbf{M}_{n\times k}(\mathbb{R})$

The vector space of *$n \times k$ matrices* ("n by k matrices") is given as follows:

$$\mathbf{M}_{n\times k}(\mathbb{R}) = \left\{ \begin{bmatrix} a_{11} & a_{12} & \ldots & a_{1k} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \ldots & a_{nk} \end{bmatrix} : a_{ij} \in \mathbb{R} \right\}$$

We write $[a_{ij}] \in \mathbf{M}_{n\times k}(\mathbb{R})$ where the bounds n and k are understood from context.

The addition on $\mathbf{M}_{n\times k}(\mathbb{R})$ is given by:

$$[a_{ij}] + [b_{ij}] = [a_{ij} + b_{ij}]$$

The scalar multiplication on $\mathbf{M}_{n\times k}(\mathbb{R})$ is given by:

$$c[a_{ij}] = [ca_{ij}]$$

The entry a_{ij} is in row i and column j.

Theorem 1.16 — Uniqueness of the Zero Vector

For any vector space V, the zero vector $\mathbf{0} \in V$ is *unique*.

Theorem 1.17 — Scaling by Zero

For any $\mathbf{z} \in V$, $0\mathbf{z} = \mathbf{0}$.

Theorem 1.18 — Additive Inverse and Scaling

For any $\mathbf{x} \in V$, $(-1)\mathbf{x} = \mathbf{x}'$.

Theorem 1.19 — Uniqueness of the Additive Inverse

Suppose that $\mathbf{v} \in V$. The additive inverse $\mathbf{v}'$ of $\mathbf{v}$ is *unique*.

Definition 1.20 — Subspaces

$W \subseteq V$ is a *subspace* of V if it is a vector space under the same operations as V.

Theorem 1.21 — Characterization of Subspaces by Closure

Let V be a vector space and $W \subseteq V$ be a non-empty subset of V. W is a subspace of V if and only if

$$c\mathbf{x} + \mathbf{y} \in W$$

for all $\mathbf{x}, \mathbf{y} \in W$ and $c \in \mathbb{F}$.

Theorem 1.22 — Subspaces Closed Under Intersection

If W_1 and W_2 are subspaces of V then $W_1 \cap W_2$ is also a subspace of V.

2 Linear Combinations

Definition 2.1 — Linear Combinations

A *linear combination* of vectors from a set $S \subseteq V$ is defined as

$$a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \ldots + a_n\mathbf{v}_n = \sum_{i=1}^{n} a_i\mathbf{v}_i$$

where $a_i \in \mathbb{F}$ and $\mathbf{v}_i \in S$. We say that $a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \ldots a_n\mathbf{v}_n$ is a linear combination of $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$.

Definition 2.2 — Spans

A span of a set of vectors S, denoted $\text{span}(S)$, is the set of all linear combinations of S. That is,

$$\text{span}(S) = \{a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \ldots + a_n\mathbf{v}_n : a_i \in \mathbb{F} \text{ and } \mathbf{v}_i \in S\}$$

For $S = \emptyset$, the span is defined as $\text{span}(\emptyset) = \{\mathbf{0}\}$.

Theorem 2.3 — Spans Are Subspaces

If $S \subseteq V$ is a subset of V, then $\text{span}(S) \subseteq V$ is a subspace of V.

Definition 2.4 — Sums of Subsets

Suppose that $A, B \subseteq V$. The *sum of subsets* is defined as

$$A + B = \{a + b : a \in A \text{ and } b \in B\}$$

Theorem 2.5 — Sums of Subspaces

If $W_1, W_2 \subseteq V$ are subspaces, then their sum $W = W_1 + W_2$ is also a subspace.

Theorem 2.6 — Sums of Subspaces

If $W_1 = \text{span}(S_1)$ and $W_2 = \text{span}(S_2)$ then $W_1 + W_2 = \text{span}(S_1 \cup S_2)$.

Definition 2.7 — Redundant Vector

A vector $\mathbf{w}$ in the set $\{\mathbf{w}, \mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is said to be *redundant* if

$$\text{span}\{\mathbf{w}, \mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\} = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$$

Definition 2.8 — Linear Dependence and Linear Independence

A set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ is *linearly dependent* if there are constants $a_i \in \mathbb{F}$ not all zero such that

$$a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \ldots + a_k\mathbf{v}_k = \mathbf{0}$$

We say that a set of vectors is *linearly independent* if it is not linearly dependent.

Theorem 2.9 — Redundant Vectors and Linear Independence

If S contains a redundant vector, then S is linearly dependent.

Note that the contrapositive is more fruitful.

If S is linearly independent, then S contains no redundant vectors.

Theorem 2.10 — Independence and Uniqueness of Linear Combinations

Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a set of vectors.

The following are equivalent:

1. S is linearly independent.
2. $\mathbf{0} \in \operatorname{span}(S)$ is represented uniquely.
3. Every vector in $\operatorname{span}(S)$ is represented uniquely.

3 Systems of Linear Equations

Definition 3.1 — Linear Systems

A *linear system* is a system of n equations in k variables $x_1, x_2, \ldots, x_k$ of the following form.

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \ldots + a_{1k}x_k & = & b_1 \\ a_{21}x_1 + a_{22}x_2 + \ldots + a_{2k}x_k & = & b_2 \\ & \vdots & \\ a_{n1}x_1 + a_{n2}x_2 + \ldots + a_{nk}x_k & = & b_n \end{cases}$$

The entries a_{ij} are *coefficients* and the values b_k are *constants*. For real vector spaces, each $a_{ij} \in \mathbb{R}$ and $b_k \in \mathbb{R}$. For complex vector spaces, each $a_{ij} \in \mathbb{C}$ and $b_k \in \mathbb{C}$.

A *solution* of the system is $(x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$ (or $(x_1, x_2, \ldots, x_n) \in \mathbb{C}^n$ when working over $\mathbb{C}$) that makes all the equations true simultaneously. Two systems are *equivalent* if they have the same set of solutions.

Remark: Linear systems can have:

1. No solutions (inconsistent)
2. One solution (independent and consistent)
3. Infinitely many solutions (dependent)

Definition 3.2 — Homogeneous Systems

A system of linear equations is *homogeneous* if all its constants are zero.

Theorem 3.3 — The Trivial Solution

A homogeneous system in k variables always admits the *trivial* solution $\mathbf{0}$.

Definition 3.4 — Augmented Matrix Notation for Linear Systems

The *augmented matrix form* of a linear system is as follows:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \ldots + a_{1k}x_k & = & b_1 \\ a_{21}x_1 + a_{22}x_2 + \ldots + a_{2k}x_k & = & b_2 \\ & \vdots & \\ a_{n1}x_1 + a_{n2}x_2 + \ldots + a_{nk}x_k & = & b_n \end{cases} \longleftrightarrow \left[\begin{array}{cccc|c} a_{11} & a_{12} & \ldots & a_{1k} & b_1 \\ a_{21} & a_{22} & \ldots & a_{2k} & b_2 \\ & & & & \vdots \\ a_{n1} & a_{n2} & \ldots & a_{nk} & b_n \end{array}\right]$$

Definition 3.5 — The Row Operations

Given a linear system, we can apply the following *elementary row operations.*

Add any multiple of one row to another row. For $i \neq j$ and any $\lambda \in \mathbb{F}$, we have:

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \dots & a_{1k} & b_1 \\ a_{21} & a_{22} & \dots & a_{2k} & b_1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{j1} & a_{j2} & \dots & a_{jk} & b_j \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nk} & b_n \end{array}\right] \xrightarrow{\lambda R_i + R_j} \left[\begin{array}{cccc|c} a_{11} & a_{12} & \dots & a_{1k} & b_1 \\ a_{21} & a_{22} & \dots & a_{2k} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \lambda a_{i1} + a_{j1} & \lambda a_{i2} + a_{j2} & \dots & \lambda a_{ik} + a_{jk} & \lambda b_i + b_j \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nk} & b_n \end{array}\right]$$

Multiply any equation by a non-zero scalar. For any $\lambda \neq 0$, we have

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \dots & a_{1k} & b_1 \\ a_{21} & a_{22} & \dots & a_{2k} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{i1} & a_{i2} & \dots & a_{ik} & b_i \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nk} & b_n \end{array}\right] \xrightarrow{\lambda R_i} \left[\begin{array}{cccc|c} a_{11} & a_{12} & \dots & a_{1k} & b_1 \\ a_{21} & a_{22} & \dots & a_{2k} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \lambda a_{i1} & \lambda a_{i2} & \dots & \lambda a_{ik} & \lambda b_i \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nk} & b_n \end{array}\right]$$

Interchange two rows. For $i \neq j$, we have:

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \dots & a_{1k} & b_1 \\ a_{21} & a_{22} & \dots & a_{2k} & b_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{i1} & a_{i2} & \dots & a_{ik} & b_i \\ \vdots & \vdots & & \vdots & \vdots \\ a_{j1} & a_{j2} & \dots & a_{jk} & b_j \\ \vdots & \vdots & & \vdots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nk} & b_n \end{array}\right] \xrightarrow{R_i \leftrightarrow R_j} \left[\begin{array}{cccc|c} a_{11} & a_{12} & \dots & a_{1k} & b_1 \\ a_{21} & a_{22} & \dots & a_{2k} & b_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{j1} & a_{j2} & \dots & a_{jk} & b_j \\ \vdots & \vdots & & \vdots & \vdots \\ a_{i1} & a_{i2} & \dots & a_{ik} & b_i \\ \vdots & \vdots & & \vdots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nk} & b_n \end{array}\right]$$

Theorem 3.6 — Row Operations Preserve the Solution Sets of Linear Systems

Applying a finite sequence of elementary row operations to a system produces an equivalent system.

Definition 3.7 — Echelon Form

A system of linear equations is in *echelon form (EF)* if the following three conditions hold:

1. In each equation EITHER all the coefficients are zero OR the first non-zero coefficient is 1. We say that first non-zero term $1x_{j(i)}$ is the *leading term* of its row. Its column is $j(i)$.
2. For each $r \neq i$, the coefficient $a_{rj(i)}$ of $x_{j(i)}$ is zero. ("Only non-zero entry in column")
3. $j(i) < j(i+1)$ whenever rows i and $i+1$ have non-zero entries. ("Echelon/staircase pattern")

Note that condition (3) causes the leading terms to form a staircase from top-left to bottom-right.

Remark: In simpler terms, three conditions in definition 3.7 are equivalent to the following:

1. In each equation, either all coefficients are zero or the first non-zero coefficient is 1.
2. If the first non-zero coefficient is 1, it is the only non-zero coefficient in its column.
3. The leading terms form a staircase from the top-left to bottom-right.

Definition 3.8 — The Elimination Algorithm

Input: An $n \times k$ matrix $M \in \mathbf{M}_{n\times k}(\mathbb{F})$. Output: A matrix M' in echelon form equivalent to M.

For each row R_i where $1 \leq i \leq n$, we proceed as follows:

Check for zero rows: If the rows R_r for $i \leq r \leq n$ are all zeroes, then we're done.

Otherwise, find the *first non-zero coefficient*, the $a_{rj} \neq 0$ with the smallest j and $r \geq i$.

- *Define the column of the leading one:* We define $j(i) = j$.
- *Divide out by a_{rj}* by applying the operation $M \xrightarrow{\frac{1}{a_{rj}}R_r} M'$
- *Exchange R_i and R_r* by applying the operation $M \xrightarrow{R_r \leftrightarrow R_i}$
- *Clear the column jth column* by applying the operation $M \xrightarrow{-a_{rj}R_i + R_j}$ for each $a_{rj} \neq 0$.

Theorem 3.9 — Every Linear System Has a Row Echelon Form

Every linear system has a *unique* row-echelon form.

Definition 3.10 — Basic and Free Variables

In the echelon form of a linear system, all variables appearing as leading terms are *basic variables*. All non-basic variables (including those with a coefficient of zero) are *free variables*.

Theorem 3.11 — Solutions of Homogeneous Systems and Parameters

If $n < k$, then every homogeneous system with k variables and n equations has a non-trivial solution.

Definition 3.12 — Affine Sets

An *affine set* is a set of the form

$$\{\mathbf{x}_p + \mathbf{v} : \mathbf{v} \in \operatorname{span}(S)\}$$

where S is a set of vectors.

4 Bases and Dimension

Definition 4.1 — Spanning Sets and Bases

A subset $S \subseteq V$ is a *spanning set* of V if $V = \text{span}(S)$.

A set $S \subseteq V$ is a *basis* of V if:

1. S is linearly independent and
2. S is a spanning set of V (i.e. $V = \text{span}(S)$)

Theorem 4.2 — Characterizing Bases by Unique Representation

S is a basis of V if and only if every vector in V has a unique representation in $\text{span}(S)$.

Lemma 4.3 — The Linear Independence Extension Lemma

Let $S \subseteq V$ be linearly independent. Suppose that $\mathbf{x} \in V$ and $\mathbf{x} \notin S$.

$S \cup \{\mathbf{x}\}$ is linearly independent if and only if $\mathbf{x} \notin \text{span}(S)$.

Theorem 4.4 — Extending a Linearly Independent Set to a Basis

Suppose that V has a finite spanning set $T = \{\mathbf{y}_1, \ldots, \mathbf{y}_n\}$ and $S = \{\mathbf{x}_1, \ldots, \mathbf{x}_k\}$ is a linearly independent subset of V. There exists a finite basis S' of V such that $S \subseteq S'$.

Remark: Theorem 4.4 can be proven by repeatedly extending S so that it remains linearly independent, as follows.

Set $S' = S$. If this is a basis, we are done. Otherwise, for each $\mathbf{y}_i$,

- If $\mathbf{y}_i \notin \text{span}(S')$, set $S' = S' \cup \{\mathbf{y}_i\}$.
- If $\mathbf{y}_i \in \text{span}(S')$, do not add $\mathbf{y}_i$, and move on to the next vector $\mathbf{y}_{i+1}$.

By the end of this process, S' will contain each $\mathbf{y}_i$. From here, we can argue that S' is linearly independent and establish that S' is also a spanning set of V (since $T \subseteq S'$).

Theorem 4.5 — The Dimension Bound

If V has a spanning set $T = \{\mathbf{y}_1, \ldots, \mathbf{y}_k\}$ and $S = \{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$ is a linearly independent set, then

$$n \leq k$$

Theorem 4.6 — All Bases are the Same Size

If $S = \{\mathbf{x}_1, \ldots, \mathbf{x}_k\}$ and $S' = \{\mathbf{y}_1, \ldots, \mathbf{y}_n\}$ are two finite bases of V, then $n = k$.

Definition 4.7 — Dimension

A vector space is *finite dimensional* if it admits a finite basis. The *dimension* $\dim(V)$ of a finite dimensional vector space is the number of elements in any basis of V.

Theorem 4.8 — Independence and Cardinality

Let V be a finite-dimensional vector space and $S \subseteq V$ be a linearly independent set. If $|S| = \dim(V)$, then S is a basis of V.

Theorem 4.9 — Proper Subspace Dimension

Let S be a proper subspace of a finite-dimensional vector space V (i.e. a subspace with $S \neq V$). It follows that $\dim(S) < \dim(V)$.

5 Linear Transformations

Definition 5.1 — Linear Transformation

Suppose that $(V, \boxplus, \boxdot)$ and $(W, \oplus, \odot)$ are two vector spaces.

A *linear transformation/map/function* is a function $T\colon V \to W$ such that:

$$T(\mathbf{u} \boxplus \mathbf{v}) = T(\mathbf{u}) \oplus T(\mathbf{v}) \qquad\qquad T(a \boxdot \mathbf{v}) = a \odot T(\mathbf{v})$$

Theorem 5.2 — Linear Transformations Preserve Additive Identities

If $T\colon V \to W$ is a linear transformation then $T(\mathbf{0}_V) = \mathbf{0}_W$.

Theorem 5.3 — Linear Transformations Preserve Addition and Scaling

$T\colon V \to W$ is a linear transformation if and only if

$$T\big((a \boxdot \mathbf{u}) \boxplus (b \boxdot \mathbf{v})\big) = \big(a \odot T(\mathbf{u})\big) \oplus \big(b \odot T(\mathbf{v})\big)$$

for all vectors $\mathbf{u}, \mathbf{v} \in V$ and scalars $a, b \in \mathbb{F}$.

Definition 5.4 — Zero Transformation

Let V and W be vector spaces. The *zero transformation* $Z\colon V \to W$ is given by

$$Z(\mathbf{v}) = \mathbf{0}_W$$

Definition 5.5 — Identity Transformation

Suppose that V is a vector space. The *identity transformation* $I\colon V \to V$ is given by

$$I(\mathbf{v}) = \mathbf{v}$$

Theorem 5.6 — Linearity of Zero and Identity Transformations

For any vector spaces V and W, the zero transformation $Z\colon V \to W$ and the identity transformation $I\colon V \to V$ are linear transformations.

Definition 5.7 — The Standard Inner Product

Suppose that $\mathbf{u} = (u_1, \ldots, u_n)$ and $\mathbf{v} = (v_1, \ldots, v_n)$ are vectors in $\mathbb{R}^n$.

The *standard inner product* of $\mathbf{u}$ and $\mathbf{v}$ is defined to be:

$$\langle \mathbf{u}, \mathbf{v} \rangle = \sum_{i=1}^{n} u_i v_i = u_1 v_1 + u_2 v_2 + \ldots + u_n v_n$$

We define the *length* or *norm* of $\mathbf{v}$ to be

$$\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle} = \sqrt{v_1^2 + v_2^2 \ldots + v_n^2}$$

Note: The inner product is linear in the sense that $\langle \mathbf{u}, a\mathbf{v} + b\mathbf{w} \rangle = a\langle \mathbf{u}, \mathbf{v} \rangle + b\langle \mathbf{u}, \mathbf{w} \rangle$.

Theorem 5.8 — Properties of the Standard Inner Product Over $\mathbb{R}^n$

Let $\mathbf{u}, \mathbf{v}, \mathbf{v}_1, \mathbf{v}_2 \in \mathbb{R}^n$ and $a \in \mathbb{R}$. The following properties hold for the standard inner product over $\mathbb{R}^n$:

1. $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$
2. $\langle \mathbf{u}, \mathbf{v}_1 + \mathbf{v}_2 \rangle = \langle \mathbf{u}, \mathbf{v}_1 \rangle + \langle \mathbf{u}, \mathbf{v}_2 \rangle$
3. $\langle \mathbf{u}, a\mathbf{v} \rangle = a\langle \mathbf{v}, \mathbf{u} \rangle$
4. $\langle \mathbf{u}, \mathbf{u} \rangle \geq 0$, where equality holds if $\mathbf{u} = \mathbf{0}$

Theorem 5.9 — Inner Products Define Angles

If $\mathbf{u}$ and $\mathbf{v}$ are non-zero vectors in $\mathbb{R}^n$, then the angle θ between $\mathbf{u}$ and $\mathbf{v}$ satisfies:

$$\cos(\theta) = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\|\|\mathbf{v}\|}$$

Theorem 5.10 — Orthogonality

Two non-zero vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ form a right angle if and only if $\langle \mathbf{u}, \mathbf{v} \rangle = 0$. We say that two such vectors are *orthogonal*.

Theorem 5.11 — Rotation in $\mathbb{R}^2$

Let $\mathbf{v} \in \mathbb{R}^2$ and $\theta \in \mathbb{R}$. The *rotation* of $\mathbf{v}$ by an angle of θ radians counter-clockwise about the origin is the linear map $R_\theta \colon \mathbb{R}^2 \to \mathbb{R}^2$ given by

$$R_\theta(x, y) = \big(x\cos(\theta) - y\sin(\theta), y\cos(\theta) + x\sin(\theta)\big)$$

Remark: When deriving the rotation map in theorem 5.11, it is useful to recall the following trigonometric angle sum identities for all $\theta, \phi \in \mathbb{R}$:

$$\sin(\phi + \theta) = \sin(\phi)\cos(\theta) + \cos(\phi)\sin(\theta)$$
$$\cos(\phi + \theta) = \cos(\phi)\cos(\theta) - \sin(\phi)\sin(\theta)$$

Theorem 5.12 — Projection in $\mathbb{R}^2$

Let $\mathbf{a} \in \mathbb{R}^2$ be a non-zero vector and $L = \operatorname{span}(\mathbf{a}) \subset \mathbb{R}^2$ be the line determined by $\mathbf{a}$. The *projection* of a vector $\mathbf{v} \in \mathbb{R}^2$ onto L is the linear transformation $P_{\mathbf{a}} \colon \mathbb{R}^2 \to \mathbb{R}^2$ given by

$$P_{\mathbf{a}}(\mathbf{v}) = \frac{\langle \mathbf{a}, \mathbf{v} \rangle}{\langle \mathbf{a}, \mathbf{a} \rangle}\mathbf{a}$$

Theorem 5.13 — Linear Maps are Determined by Their Actions on a Basis

Suppose that V is a finite dimensional vector space with basis $\alpha = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$. If $S \colon V \to W$ and $T \colon V \to W$ are linear maps such that $S(\mathbf{v}_i) = T(\mathbf{v}_i)$ for each $\mathbf{v}_i$, then $S(\mathbf{v}) = T(\mathbf{v})$ for all $\mathbf{v} \in V$.

Theorem 5.14 — $T \colon V \to W$ Characterized by $\dim(V) \times \dim(W)$ Numbers

Let V and W be finite dimensional vector spaces with $\dim(V) = n$ and $\dim(W) = k$. Any linear transformation $T \colon V \to W$ is uniquely determined by nk scalars.

Definition 5.15 — The Matrix $[T]_\alpha^\beta$ of a Linear Transformation

Let V and W be finite dimensional vector spaces with $\dim(V) = n$ and $\dim(W) = k$. Fix bases $\alpha = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ and $\beta = \{\mathbf{w}_1, \ldots, \mathbf{w}_k\}$ with $V = \text{span}(\alpha)$ and $W = \text{span}(\beta)$. Suppose that $T \colon V \to W$ is a linear transformation with

$$T(\mathbf{v}_i) = a_{1i}\mathbf{w}_1 + a_{2i}\mathbf{w}_2 + \ldots + a_{ki}\mathbf{w}_k$$

The *matrix of T with respect to α and β* is

$$[T]_\alpha^\beta = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{k1} & a_{k2} & \cdots & a_{kn} \end{bmatrix}$$

Note: The coefficients in $T(\mathbf{v}_i) = a_{1i}\mathbf{w}_1 + a_{2i}\mathbf{w}_2 + \ldots + a_{ki}\mathbf{w}_k$ occur as the ith column of $[T]_\alpha^\beta$.

Definition 5.16 — Coordinate Vectors

Let $\beta = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ be a basis of V. Any vector in $\mathbf{v} \in V$ can be written uniquely as

$$v = b_1\mathbf{v}_1 + b_2\mathbf{v}_2 + \ldots + b_n\mathbf{v}_n$$

We define the *coordinate vector of* $\mathbf{v}$ *with respect to* β to be:

$$[\mathbf{v}]_\beta = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

Definition 5.17 — Matrix-Vector Multiplication

Given a matrix $[T]_\alpha^\beta$ and column vector $[\mathbf{v}]_\alpha$ we define the *product* of $[T]_\alpha^\beta$ with $[\mathbf{v}]_\alpha$ as follows.

$$[T]_\alpha^\beta[\mathbf{v}]_\alpha = \begin{bmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{k1} & a_{k2} & \ldots & a_{kn} \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} = \begin{bmatrix} a_{11}b_1 + a_{12}b_2 + \ldots + a_{1n}b_n \\ a_{21}b_1 + a_{22}b_2 + \ldots + a_{2n}b_n \\ \vdots \\ a_{k1}b_1 + a_{k2}b_2 + \ldots + a_{kn}b_n \end{bmatrix}$$

Note: We multiply each row of $[T]_\alpha^\beta$ on to the column $[\mathbf{v}]_\alpha$.

Also, the product has size "rows $\times$ columns" given by $(k \times n)(n \times 1) = k \times 1$.

Theorem 5.18 — Matrix-Vector Multiplication in Coordinates $[T(\mathbf{v})]_\beta = [T]_\alpha^\beta[\mathbf{v}]_\alpha$

Fix bases $\alpha = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ of V and $\beta = \{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_k\}$ of W. If $T \colon V \to W$ is a linear transformation, then:

$$[T(\mathbf{v})]_\beta = [T]_\alpha^\beta[\mathbf{v}]_\alpha$$

Definition 5.19 — The Transformation Defined by a Matrix

Fix bases $\alpha = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ of V and $\beta = \{\mathbf{w}_1, \dots, \mathbf{w}_k\}$ of W. If $M \in \mathbf{M}_{k\times n}(\mathbb{R})$, then the transformation defined by M is $T_M \colon V \to W$ such that:

$$T_M(\mathbf{v}) = \mathbf{w} \iff M[\mathbf{v}]_\alpha = [\mathbf{w}]_\beta$$

Theorem 5.20 — The Fundamental Theorem of Matrices

1. If $M \in \mathbf{M}_{n\times k}(\mathbb{R})$, then T_M is a linear transformation.
2. If $T \colon V \to W$ has matrix $M = [T]_\alpha^\beta$ then $T = T_M$.
3. If $M \in \mathbf{M}_{n\times k}(\mathbb{R})$, then $M = [T_M]_\alpha^\beta$.

Remark: Informally, theorem 5.20 states that linear transformations between finite dimensional vector spaces behave exactly like matrices.

6 Kernel and Image

Definition 6.1 — Kernel

The *kernel* (or *nullspace*) of a linear map $T\colon V \to W$ is:

$$\ker(T) = \{\mathbf{v} \in V : T(\mathbf{v}) = \mathbf{0}_W\}$$

Theorem 6.2 — The Kernel is a Subspace

For any linear map $T\colon V \to W$, the kernel $\ker(T)$ is a subspace of V.

Definition 6.3 — Image

The *image* of a linear map $T\colon V \to W$ is:

$$\mathrm{image}(T) = \{\mathbf{w} \in W : \mathbf{w} = T(\mathbf{v}) \text{ for some } \mathbf{v} \in V\}$$

Theorem 6.4 — The Image is a Subspace

For any linear map $T\colon V \to W$, the image $\mathrm{image}(T)$ is a subspace of W.

Theorem 6.5 — Kernels in Coordinates

Kernels are solutions of homogeneous linear systems.

$$T(\mathbf{x}) = \mathbf{0} \iff [T]_\alpha^\beta[\mathbf{x}]_\alpha = [\mathbf{0}]_\beta$$

Theorem 6.6 — Image and Spans

If $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is a spanning set for V and $T\colon V \to W$ is a linear transformation, then $\{T(\mathbf{v}_1), \ldots, T(\mathbf{v}_n)\}$ is a spanning set for $\mathrm{image}(T)$.

Theorem 6.7 — Image and Columns ("Procedure 1")

Fix bases $\alpha = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ and $\beta = \{\mathbf{w}_1, \ldots, \mathbf{w}_k\}$ of V and W. The image $\mathrm{image}(T)$ is spanned by the columns of $[T]_\alpha^\beta$ which contain basic variables of the homogeneous system $[T]_\alpha^\beta[\mathbf{v}]_\alpha = \mathbf{0}$.

Theorem 6.8 — Bases for Image and Kernel ("Procedure 2")

If V and W are finite dimensional vector spaces and $\dim(V) = n$ then there is a basis:

$$V = \mathrm{span}\{\mathbf{u_1}, \mathbf{u}_2, \ldots, \mathbf{u}_k, \mathbf{v}_{k+1}, \ldots, \mathbf{v}_n\}$$

such that: $\ker(T) = \mathrm{span}\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k\}$ and $\mathrm{image}(T) = \mathrm{span}\{T(\mathbf{v}_{k+1}), \ldots, T(\mathbf{v}_n)\}$.

Theorem 6.9 — The Rank-Nullity Theorem

If V and W are finite dimensional vector spaces and $T\colon V \to W$ is linear, then:

$$\dim\big(\ker(T)\big) + \dim\big(\mathrm{image}(T)\big) = \dim(V)$$

7 Applications of the Dimension Theorem

Definition 7.1 — Injective and Surjective

A linear transformation $T\colon V \to W$ is *injective/one-to-one* if:

$$T(\mathbf{x}) = T(\mathbf{y}) \implies \mathbf{x} = \mathbf{y}$$

A linear transformation $T\colon V \to W$ is *surjective/onto* if:

$$\text{for all } \mathbf{w} \in W \text{ there is } \mathbf{v} \in V \text{ such that } T(\mathbf{v}) = \mathbf{w}$$

Theorem 7.2 — Kernels and Injectivity

Let $T\colon V \to W$ be a linear transformation. The following are equivalent:

1. T is injective.
2. $\ker(T) = \{\mathbf{0}_V\}$
3. $\dim\big(\ker(T)\big) = 0$

Theorem 7.3 — Image and Surjectivity

Let $T\colon V \to W$ be a linear transformation, where W is a finite dimensional vector space. The following are equivalent:

1. T is surjective.
2. $\text{image}(T) = W$
3. $\dim\big(\text{image}(T)\big) = \dim(W)$

Theorem 7.4 — Dimension and Injectivity

Suppose that V and W are finite dimensional vector spaces. If $\dim(W) < \dim(V)$ then $T\colon V \to W$ is not injective.

Note: The contrapositive is also helpful: If $T\colon V \to W$ is injective then $\dim(V) \leq \dim(W)$.

Theorem 7.5 — Dimension and Surjectivity

Suppose that V and W are finite dimensional vector spaces. If $\dim(V) < \dim(W)$ then $T\colon V \to W$ is not surjective.

Note: The contrapositive is also helpful: If $T\colon V \to W$ is surjective the $\dim(W) \leq \dim(V)$.

Theorem 7.6 — In Finite Dimensions Injectivity and Surjectivity are Equivalent

Suppose that $T\colon V \to W$ and $\dim(V) = \dim(W)$, where both dimensions are finite. T is injective if and only if T is surjective.

Definition 7.7 — Inverse Sets

Suppose $T\colon V \to W$. The *inverse image* of S is: $T^{-1}(S) = \{\mathbf{v} \in V : T(\mathbf{v}) \in S\}$.

Note: This notation is weird. The function T might not have a real inverse function $T^{-1}\colon W \to V$.

Definition 7.8 — Affine Subsets

An *affine subset* of V is:

$$\mathbf{v} + S = \{\mathbf{v} + \mathbf{s} \colon \mathbf{s} \in S\}$$

for some subspace $S \subseteq V$.

Note: An affine subset is like a subspace, but it has been shifted by a vector. We use the notation

$$\mathbf{v} + S = \{\mathbf{v}\} + S$$

Theorem 7.9 — Solutions Sets of Linear Systems

The solution set S of $A\mathbf{x} = \mathbf{b}$ is an affine subset $\mathbf{x}_p + \ker(T_A)$. Formally, all solutions can be written

$$\mathbf{x} = \mathbf{x}_p + \mathbf{x}_h$$

where $\mathbf{x}_p$ is a particular solution such that $A\mathbf{x}_p = \mathbf{b}$ and $\mathbf{x}_h$ is a homogeneous solution of $A\mathbf{x}_h = \mathbf{0}$.

Theorem 7.10 — Solutions Sets of Linear Systems and Kernels

The solution set S of $A\mathbf{x} = \mathbf{b}$ is a subspace if and only if $\mathbf{b} = \mathbf{0}$.

Note: If S is a subspace then it is a kernel $\ker(T_A)$.

8 Composition and Inverses

Definition 8.1 — Composition

Given two functions $T\colon U \to V$ and $S\colon V \to W$ we define their *composition* to be: $ST\colon U \to W$ by $ST(\mathbf{x}) = S(T(\mathbf{x}))$. We also use the notation $ST = S \circ T$. We read the notation $S \circ T$ as "S of T".

Note: For a composition to be defined, the inputs and outputs must match according to the following:

$$U \xrightarrow{T} V \xrightarrow{S} W$$

Observe that $\text{image}(T) \subseteq \text{domain}(S)$.

Theorem 8.2 — A Composition of Linear Maps is Linear

If $T\colon U \to V$ and $S\colon V \to W$ are linear then $ST\colon U \to W$ is also linear.

Definition 8.3 — Addition of Linear Maps

Suppose that $T\colon V \to W$ and $S\colon V \to W$ are linear transformations. We define their *sum* as:

$$(S + T)(\mathbf{v}) = S(\mathbf{v}) + T(\mathbf{v})$$

Theorem 8.4 — Composition Has Nice Properties

1. Composition is associative: If we have three linear maps

$$R\colon U \to V \qquad S\colon V \to W \qquad T\colon W \to X$$

   then $(TS)R = T(SR)$.

2. Composition is distributive: If we have three linear maps:

$$R\colon U \to V \qquad S\colon V \to W \qquad T\colon V \to W$$

   then $(T + S)R = TR + SR$.

Note: These conditions look complicated and hard to memorize, but they're necessary to ensure that the compositions are defined. These conditions ensure there are no comp. sci. style "type errors".

Theorem 8.5 — Kernels, Images, and Composition

Suppose $S\colon U \to V$ and $T\colon V \to W$ are linear maps:

1. $\ker(S) \subseteq \ker(TS)$
2. $\text{image}(TS) \subseteq \text{image}(T)$

Definition 8.6 — The Matrix Product

Fix bases $\alpha = \{\mathbf{u}_1, \ldots, \mathbf{u}_n\}$, $\beta = \{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$, $\gamma = \{\mathbf{w}_1, \ldots, \mathbf{w}_\ell\}$ of U, V, and W respectively.

If $S\colon U \to V$ and $T\colon V \to W$ then we define the *product* of $[T]_\beta^\gamma$ and $[S]_\alpha^\beta$ as follows:

$$[T]_\beta^\gamma [S]_\alpha^\beta = [TS]_\alpha^\gamma$$

Theorem 8.7 — The Matrix Multiplication Formula

If $T = [t_{ij}] \in \mathbf{M}_{\ell\times k}(\mathbb{F})$ and $S = [s_{ij}] \in \mathbf{M}_{k\times n}(\mathbb{F})$ then this gives: $TS = [p_{ij}]$ where:

$$p_{ij} = \sum_{N=1}^{k} t_{iN} s_{Nj}$$

Definition 8.8 — An Inverse of a Function

Suppose that $S\colon V \to W$ is a function. An *inverse* of S is $T\colon W \to V$ such that:

$$TS(\mathbf{v}) = \mathbf{v} \text{ for all } \mathbf{v} \in V \text{ and } ST(\mathbf{w}) = \mathbf{w} \text{ for all } \mathbf{w} \in W$$

Theorem 8.9 — Injectivity, Surjectivity, and Inverses

A function $S\colon V \to W$ has an inverse $T\colon W \to V$ if and only if S is injective and surjective.

Theorem 8.10 — Inverses are Unique

If a function $S\colon V \to W$ has an inverse $T\colon W \to V$ then T is unique.

Theorem 8.11 — The Inverse is Linear

If a linear transformation $S\colon V \to W$ has an inverse $T\colon W \to V$ then T is linear.

Theorem 8.12 — Elementary Row Operations are Invertible Matrices

Each elementary row operation $M \xrightarrow{X} M'$ on an $n \times n$ matrix M can be represented by an invertible matrix E_X such that: $M' = E_X M$.

Definition 8.13 — Isomorphism

An invertible linear transformation $T\colon V \to W$ is an *isomorphism*.

Definition 8.14 — Endomorphism

A linear transformation $T\colon V \to V$ is an *endomorphism*.

Definition 8.15 — Automorphism

An invertible linear transformation $T\colon V \to V$ is an *automorphism*.

Definition 8.16 — Sets of Linear Transformations

The set $\mathcal{L}(V, W)$ is the set of all linear transformations from V to W.

The set $\mathcal{L}(V, V)$ (also denoted by $\mathcal{L}(V)$) is the set of all endomrophisms mapping V to itself.

The set $\mathrm{Aut}(V) \subset \mathcal{L}(V, V)$ is the set of invertible linear maps $T\colon V \to V$.

Remark: In this class, we have extensively studied vector spaces as examples of algebraic structures. We've studied them from the axioms and built up an elaborate theory of how they work. However, there are other kinds of algebraic structures worth studying. Groups are algebraic structures with a very simple set of axioms:

1. There is an associative multiplication operation $G \times G \to G$.
2. There is an identity element $e \in G$ such that: $eg = ge = g$.
3. For any $g \in G$ there is an inverse $g^{-1} \in G$ such that $gg^{-1} = g^{-1}g = e$.

Theorem 8.17 — The Automorphism Group of a Vector Space

Let V be a vector space. The set $\mathrm{Aut}(V) \subset \mathcal{L}(V, V)$ is a group.

Theorem 8.18 — The Gauss-Jordan Algorithm for Finding Inverses

Input: A matrix $M \in \mathbf{M}_{n\times n}(\mathbb{F})$. Output: If M has an inverse, output M^{-1}.

1. Form the augmented matrix $[M|I]$ with an $n \times n$ identity.
2. Reduce M in $[M|I]$ to EF and obtain $[A|M']$.
3. If $A = I$ then $M' = M^{-1}$. Otherwise, M is not invertible.

Theorem 8.19 — Isomorphisms and Injectivity/Surjectivity

A linear transformation $T \colon V \to W$ is an isomorphism if and only if it is injective and surjective.

Note: If $\dim(V) = \dim(W)$ are finite, then it suffices to check either injective or surjective.

Definition 8.20 — Isomorphic

We say that V and W are *isomorphic* if there exists an isomorphism $T \colon V \to W$.

Theorem 8.21 — The Classification of Finite Dimensional Vector Spaces

If V is an n-dimensional vector space over the field $\mathbb{F}$, then V is isomorphic to $\mathbb{F}^n$.

Corollary 8.22 — Real Vector Spaces Isomorphism and Dimensions

Two finite dimensional real vector spaces V and W are isomorphic if and only if $\dim(V) = \dim(W)$.

9 Change of Basis and Determinants

Definition 9.1 — Alternating and Multilinear

A function $T\colon V^n \to W$ is *multilinear* if: it is linear in each of its n inputs.

$$T(\mathbf{v}_1, \mathbf{v}_2, \dots, a\mathbf{v}_i + b\mathbf{v}_i', \dots, \mathbf{v}_n) = aT(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_i, \dots, \mathbf{v}_n) + bT(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_i', \dots, \mathbf{v}_n)$$

A function $T\colon V^n \to \mathbb{R}$ is *alternating* if: switching the order of two entries changes the sign.

$$T(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_i, \dots, \mathbf{v}_j, \dots, \mathbf{v}_n) = -T(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_j, \dots, \mathbf{v}_i, \dots, \mathbf{v}_n)$$

Theorem 9.2 — Alternating and Repeated Entries

If the function $T\colon V^n \to \mathbb{R}$ is alternating and has two identical inputs, then it outputs zero.

$$T(\mathbf{v}_1, \dots, \mathbf{v}_i, \dots, \mathbf{v}_i, \dots, \mathbf{v}_n) = 0$$

Lemma 9.3 — Alternating, Multilinear, and Dependent Input Set

Consider a function $T\colon V^n \to W$. If T is alternating and multilinear, and the set of input vectors are linearly dependent, then the output will be zero.

Theorem 9.4 — The Characterization of the det Function

Consider $\mathbf{M}_{n\times n}(\mathbb{R})$ as $(\mathbb{R}^n)^n$ where the rows of the matrix are thought of as vectors in $\mathbb{R}^n$. If $f\colon \mathbf{M}_{n\times n}(\mathbb{R}) \to \mathbb{R}$ is alternating, multilinear, and satisfies $f(I) = 1$, then $f = \det$.

Note: The proof of this theorem is not required for this course.

Theorem 9.5 — Diagonal Matrices

A matrix $D = [d_{ij}] \in \mathbf{M}_{n\times n}(\mathbb{R})$ is *diagonal* if $d_{ij} = 0$ when $i \neq j$.

The determinant of a diagonal matrix is: $\det(D) = d_{11}d_{22}\dots d_{nn}$.

Theorem 9.6 — Determinants and Row Operations

Let $M \in \mathbf{M}_{n\times n}(\mathbb{R})$. Suppose that M' is obtained from M by row operation.

Scale a row. If $M \xrightarrow{\lambda R_i} M'$ then $\det(M') = \lambda \det(M)$.

Exchange R_i and R_j. If $M \xrightarrow{R_i \leftrightarrow R_j} M'$ then $\det(M') = -\det(M)$.

Add a multiple of a row. If $M \xrightarrow{\lambda R_i + R_j} M'$ then $\det(M') = \det(M)$.

Note: These properties of det are sometimes called “row operation invariance”. This is not quite right, because the determinant does change when applying row-operations. However, the determinant changes in simple and predictable ways.

Theorem 9.7 — Determinants and Invertibility

A matrix $M \in \mathbf{M}_{n\times n}(\mathbb{R})$ is invertible if and only if $\det(M) \neq 0$.

Definition 9.8 — The 1×1 and 2×2 Determinants

For a 1×1 matrix, we have the following formula: $\det([a]) = a$.

For a 2×2 matrix, we have the following formula:

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$$

Definition 9.9 — Minors (Submatrices)

The *ij-minor* of $M \in \mathbf{M}_{n \times n}(\mathbb{R})$ is the matrix M_{ij} obtained by deleting row i and column j from M.

Note: The ij-minor $M_{ij} \in \mathbf{M}_{(n-1)\times(n-1)}(\mathbb{R})$ is a smaller matrix than M.

Theorem 9.10 — The Recursive Formula for $\det$

If $M = [m_{ij}] \in \mathbf{M}_{n \times n}(\mathbb{R})$ and $1 \leq i \leq n$ then we define:

$$\det(M) = \sum_{j=1}^{n} (-1)^{i+j} m_{ij} \det(M_{ij})$$

or

$$\det(M) = \sum_{i=1}^{n} (-1)^{i+j} m_{ij} \det(M_{ij})$$

This definition produces the unique alternating and multilinear function such that $\det(I) = 1$.

Theorem 9.11 — Changing Coordinates

Suppose that V is a finite dimensional vector space with bases α and β. Let $I : V \to V$ be the identity transformation. The coordinate vectors of $[\mathbf{v}]_\alpha$ and $[\mathbf{v}]_\beta$ are related as follows:

$$[I]_\alpha^\beta [\mathbf{v}]_\alpha = [\mathbf{v}]_\beta$$

We define the *change of basis* matrix from α to β as $[I]_\alpha^\beta$.

Theorem 9.12 — Changing Coordinates and Linear Maps

Suppose that $T : V \to W$ is a linear map given as a matrix $[T]_\alpha^\beta$. If V has two bases α and α' and W has two bases β and β' then:

$$[T]_{\alpha'}^{\beta'} = [I_W]_\beta^{\beta'} [T]_\alpha^\beta [I_V]_{\alpha'}^\alpha$$

Theorem 9.13 — A Useful Special Case: Changing Coordinates of $T : V \to V$

Suppose that $T : V \to V$ is a linear map given as a matrix $[T]_\alpha^\alpha$. If V has two bases α and α', then

$$[T]_{\alpha'}^{\alpha'} = [I]_\alpha^{\alpha'} [T]_\alpha^\alpha [I]_{\alpha'}^\alpha = \left([I]_{\alpha'}^\alpha\right)^{-1} [T]_\alpha^\alpha [I]_{\alpha'}^\alpha$$

10 Eigenvalues and Eigenvectors

Definition 10.1 — n-fold Composition

If we have a linear transformation $T: V \to V$ we can consider applying it repeatedly. We write

$$T^n(\mathbf{v}) = T\Big(T\Big(\dots(T(\mathbf{v}))\Big)\Big)$$

where T is applied n times. This defines the *n-fold composition* of T.

Definition 10.2 — Eigenvalues and Eigenvectors

Suppose that a linear map $T: V \to V$ satisfies: $T(\mathbf{x}) = \lambda \mathbf{x}$ for some non-zero $\mathbf{x} \in V$.

We say that $\mathbf{x}$ is an *eigenvector* with *eigenvalue* λ. (German: "characteristic vector / value")

Theorem 10.3 — Eigenvectors Are Vectors in Kernels

$$T(\mathbf{v}) = \lambda \mathbf{v} \iff \mathbf{v} \in \ker(T - \lambda I)$$

Definition 10.4 — The Characteristic Polynomial

The *characteristic polynomial* of a matrix A is $\chi_A(\lambda) = \det(A - \lambda I)$.

Theorem 10.5 — Eigenvalues are Roots of the Characteristic Polynomial

A scalar λ is an eigenvalue of T if and only if it is a root of $\chi_{[T]_\alpha}$.

Definition 10.6 — The λ-Eigenspace E_λ

If $T: V \to V$ is a linear transformation then its *λ-eigenspace* is: $E_\lambda = \{\mathbf{v} : T(\mathbf{v}) = \lambda \mathbf{v}\}$.

Theorem 10.7 — E_λ is a Subspace

For any $\lambda \in \mathbb{R}$, we have that E_λ is a subspace of V.

Theorem 10.8 — Determinants and Eigenvalues

If A has eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ then $\det(A) = \prod_{i=1}^{n} \lambda_i$.

Theorem 10.9 — Invertibility and Eigenvalues

A is invertible if and only if $\lambda = 0$ is not an eigenvalue.

11 Diagonalizability

Theorem 11.1 — The LADR Approach to Finding Eigenvectors: Find a Dependence

Suppose that V is an n-dimensional real vector space. If $T\colon V \to V$ is a linear transformation, then $\{I, T^1, \ldots, T^{n\cdot n}\}$ is linearly dependent in $\mathcal{L}(V, V)$. Alternatively, there are non-zero constants such that:

$$Z = a_{n\cdot n}T^{n\cdot n} + \ldots + a_2T^2 + a_1T^1 + a_0I$$

where $I\colon V \to V$ is the identity and $Z\colon V \to V$ is the zero map.

Theorem 11.2 — The LADR Approach: Factor and Apply the Dependence

Suppose that $T\colon V \to V$ is linear map of an n-dimensional real vector space. Moreover, suppose that $p(T) = Z$ for some polynomial $p \in \mathbf{P}_n(\mathbb{R})$ that factors as:

$$Z = T^{n\cdot n} + \ldots + a_2T^2 + a_1T^1 + a_0I = (T - \lambda_kI)(T - \lambda_{k-1}I)\ldots(T - \lambda_1I)$$

For any non-zero vector $\mathbf{v} \in V$ we have that one of the vectors

$$\{\underbrace{(T-\lambda_1 I)}_{\text{One term}}\mathbf{v}, \underbrace{(T-\lambda_2 I)(T-\lambda_1 I)}_{\text{Two terms}}\mathbf{v}, \ldots, \underbrace{(T-\lambda_{k-1} I)\ldots(T-\lambda_1 I)}_{k-1 \text{ terms}}\mathbf{v}\}$$

is an eigenvector of T.

Theorem 11.3 — The Fundamental Theorem of Algebra

If $p(x) \in \mathbf{P}_n(\mathbb{R})$ is a degree n polynomial with real coefficients of the following form:

$$p(x) = x^n + a_{n-1}x^{n-1} + \ldots + a_0$$

then $p(x)$ factors uniquely in the following form:

$$p(x) = (x - \lambda_1)^{m_1}(x - \lambda_2)^{m_2}\ldots(x - \lambda_k)^{m_k}$$

where $m_1 + m_2 + \ldots + m_k = n$ and $\lambda_i \in \mathbf{C}$ are complex numbers.

Note: For us, everything will factor completely over the reals and so $\lambda_i \in \mathbb{R}$.

Theorem 11.4 — A Bound on The Number of Eigenvalues

If $M \in \mathbf{M}_{n\times n}(\mathbb{R})$ then M has at most n real eigenvalues.

Definition 11.5 — Diagonalizable

A linear transformation $T\colon V \to V$ of a finite-dimensional vector space is *diagonalizable* if there is a basis α of V consisting of eigenvectors of T.

Theorem 11.6 — Characterization of Diagonalizable

$T\colon V \to V$ is diagonalizable if and only if $[T]_\alpha^\alpha$ is similar to a diagonal matrix.

Theorem 11.7 — Eigenvectors of Distinct Eigenvalues are Independent

Suppose that $T(\mathbf{v}_i) = \lambda_i\mathbf{v}_i$ for $i = 1, 2, \ldots, n$ are distinct eigenvalues and eigenvectors of T. The set of vectors $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is linearly independent.

Theorem 11.8 — The Nice Case of Diagonalization

Suppose that V is an n-dimensional real vector space.

If $T\colon V \to V$ has n distinct real eigenvalues, then T is diagonalizable.

Definition 11.9 — Algebraic and Geometric Multiplicity

The *algebraic multiplicity* $\mathrm{Alg}(\lambda)$ of an eigenvalue λ is the number of times that it occurs as a root of the characteristic polynomial of T. The *geometric multiplicity* $\mathrm{Geo}(\lambda)$ of an eigenvalue λ is the dimension $\dim(E_\lambda)$.

Theorem 11.10 — Geometry $\leq$ Algebra

Suppose that V is an n-dimensional real vector space.

If $T\colon V \to V$ has a real eigenvalue λ then $\mathrm{Geo}(\lambda) \leq \mathrm{Alg}(\lambda)$.

Theorem 11.11 — Independent Sets from Eigenspaces

Suppose that V is an n-dimensional real vector space with distinct real eigenvalues λ_1 and λ_2. If $S_1 \subset E_{\lambda_1}$ and $S_2 \subset E_{\lambda_2}$ are sets of independent vectors, then $S_1 \cup S_2$ is independent.

Theorem 11.12 — The General Case of Diagonalization

Suppose that $T\colon V \to V$ is a linear transformation of a finite dimensional vector space. Furthermore, suppose that it has distinct real eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_k$. Let m_i be the algebraic multiplicity of the eigenvalue λ_i.

T is diagonalizable if and only if:

1. $\dim(V) = m_1 + m_2 + \ldots + m_k$
2. $\dim(E_{\lambda_i}) = m_i$